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Who does well after a stroke? The Sydney Stroke Study

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Research addressing positive outcomes one year after stroke has been limited. The sample comprised 125 participants with complete Activities of Daily Living (ADL), Instrumental Activities of Daily Living (IADL) and Mini-Mental State Examination (MMSE) scale scores at baseline (~4 months after ischaemic stroke) and at follow-up (1 year later), 31 persons were defined as having a favourable outcome (an MMSE score of ≥28/30 and combined ADL/IADL score equal to 14/14 at follow-up) and 94 as having a poorer outcome. Predictors of a favourable outcome following stroke included being younger, having higher premorbid IQ, no atrial fibrillation, no dementia, less apathy and fewer intercurrent cerebrovascular events. We conclude that people can have good outcomes in the year after stroke except if they experience further cerebrovascular events and/or have risk factors for cerebrovascular disease. Brain reserve appears to be protective.

Keywords: stroke; cognition; function; vascular; outcome assessment

Introduction

There is little research that investigates the correlates of positive outcomes in the year following stroke. The majority of work in this area has explored predictors of disability after stroke as measured by limitations in Activities of Daily Living (ADL) and/or physical function. A review of studies examining functional outcomes at varying time points post-stroke identified age, level of consciousness 48 h after stroke, disability at admission, urinary incontinence, extent of motor paresis, sitting balance, status after recurrent stroke, orientation to time and place and perceived social support as predictors of functional recovery (Kwakkel, Wagenaar, Kollen, & Lankhorst, 1996). Most of these studies had evaluated performance in the period immediately following the stroke. Other variables associated with poorer functional recovery three months after stroke include diagnosis of depression (Gainotti, Antonucci, Marra, & Paolucci, 2001), hypertension, larger infarct, internal carotid artery territory infarct and lower neurological score at admission (Paithankar & Dabhi, 2003).

There have also been relatively few studies that have considered outcomes with respect to cognitive function. Kotila, Waltimo, Niemi, Laaksonen, and Lempinen (1984) examined people 3–12 months post-stroke and found significant improvement across neurological, neuropsychological and functional domains. Desmond, Moroney, Sano, and Stern (1996) examined 151 participants with ischaemic stroke (mean age: 70.4 ± 7.7 years) at 3 months and at 12–36 months after stroke. Only 19 persons, i.e., 12.6%, met the study's rigorous criterion for improvement: an increase in the cognitive summary score greater than the two standard deviations (SDs) above the average changes of the control group. Compared to the remainder of the group with stroke, improvers were more likely to have had a left hemispheric infarct compared to brain stem/cerebellar infarct, exhibit neglect, have a lower incidence of diabetes and be significantly more impaired cognitively at baseline (i.e., have greater possibility for improvement).

Ballard, Rowan, Stephens, Kalaria, and Kenny (2003) evaluated cognitive function in an older cohort of 115 people (mean age: 80.4 ± 3.8 years), 3 and 15 months after their stroke. A baseline diagnosis of dementia led to exclusion. Improvement was defined as a greater than two-point increase in MMSE (Folstein, Folstein, & McHugh, 1975) score, decline was classified as a diagnosis of incident dementia and the remainder were classed as stable. Fifty-seven participants (i.e., 50%) had some gain in global cognition between the two assessments, although only 18 of these met the criterion for improvement; and 10 persons were diagnosed with dementia. Changes in depression or delirium were excluded as explanations for improvement. Compared to the stable group at baseline, those who improved had significantly lower scores on attention, orientation and total score on the Cambridge Assessment of Mental Disorders in the Elderly, section B (CAMCOG; Roth et al., 1986). Those with diabetes were significantly less likely to be improvers.

In summary, there have been few long-term follow-up studies investigating positive functional and cognitive outcomes in stroke survivors and no single study

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has defined outcome as including both physical and cognitive performance. Most of the functional outcome studies were conducted in the period immediately after stroke and the cognitive outcome studies amalgamated non-improvers even though this group is heterogeneous comprising decliners, those who were stable and those performing at ceiling, who by definition cannot improve. Given the limited research evaluating favourable outcomes following stroke, the aim of this study was to examine longitudinally the clinical characteristics of a group who do well after stroke as opposed to those with poorer outcomes. We examined a broader concept of favourable outcome defined according to cognitive and functional measures and which includes improvers as well as those continuing to perform at a high level.

Methods

Subjects

Consecutive eligible persons hospitalised for ischaemic stroke were asked to participate (see procedure further). An ischaemic stroke was defined as 'rapidly developing clinical signs of focal (or global) disturbance of cerebral function, with symptoms lasting 24 h or longer, with no apparent cause other than of vascular origin' in which a brain Computerised Tomography (CT) or Magnetic Resonance Imaging (MRI) scan does not show intracranial haemorrhage (Kunitz et al., 1984). Exclusion criteria were: age over 85 years; haemorrhagic stroke; transient ischaemic attack; impairment of consciousness persisting for greater than 7 days at time of stroke; inability to give informed consent; insufficient fluency in English to complete testing; past history of dementia; other neurological diseases known to affect cognition; current alcohol or drug abuse; a retrospective score on the Informant Questionnaire on Cognitive Decline in the Elderly (IQCODE; Jorm, 1994) $\geq$3.31 (indicating possible dementia prior to the index stroke); a Diagnostic and Statistical Manual of Mental Disorders 4th Edition (DSM-IV; American Psychiatric Association, 1994); diagnosis of mental retardation; severe aphasia (<3 on the Aphasia Severity Rating Scale of the Boston Diagnostic Aphasia Examination; Goodglass & Kaplan, 1983); and lack of an appropriate informant.

From consecutive admissions over a 38-month period between May 1997 and June 2000 to the inpatient stroke units at Prince of Wales and St George Hospitals in Sydney, Australia, 205 people met inclusion and exclusion criteria and gave consent for entry into the study. By the time of detailed baseline assessment, 3–6 months following the cerebrovascular event, 37 were lost to follow-up (27 due to withdrawal, three died, four relocated outside Sydney, two were uncontactable and one was excluded for baseline dementia/aphasia missed earlier). The remaining 168 eligible people were scheduled for medical/psychiatric and neuropsychological assessments at baseline and at 12–15 months follow-up. The sample for this study comprised the 125 participants who had complete ADL, IADL and MMSE scales at both time-points. The study had approval of *institutional ethics committee* and subjects provided written informed consent after receiving a complete description of the study.

'Doing well' definitions

A favourable outcome was defined as having an MMSE score $\geq$28/30 in addition to an ADL + IADL score equal to 14/14 at the follow-up assessment. This definition included improvers as well as those with stable scores. The group defined as having a poorer outcome after stroke comprised those with an MMSE <28 and/or ADL + IADL <14 at follow-up. The cut-off for the MMSE score was chosen according to the guidelines of Crum, Anthony, Bassett, and Folstein (1993). The cut-off for combined ADL + IADL score was calculated according to the mean of the control group at follow-up, which was 13.81 (0.48). The control subjects ($n = 106$ at baseline) were recruited by advertisement and via local community groups. Control subjects were excluded if they had a history of stroke or any of the above exclusion criteria. The control and stroke participant groups were matched for age and gender.

Clinical measures

Stroke diagnoses were made according to The International Statistical Classification of Diseases and Related Health Problems, Tenth Revision (ICD-10; World Health Organization, 1992) criteria and *were* confirmed by consensus at team clinical meetings. Severity was established using the European Stroke Scale (ESS; Hantson et al., 1994). Functional ability was measured with the ADL (Katz & Akpom, 1976) and IADL (Lawton & Brody, 1969) scales. Items were rated as independent (1) or at least some dependency (0) with the total score being the sum of the number of items *independently performed* (maximum score = 14). Scores were pro-rated for items marked as non-applicable. Vascular risk factors (i.e., hypertension, diabetes mellitus, myocardial infarction, angina, coronary artery disease, atrial fibrillation, hypercholesterolemia and smoking) and current and past alcohol use were recorded.

Psychiatric measures

Psychiatric measures included the Structured Clinical Interview for DSM-IV (SCID-I; First, Spitzer, Gibbon, & Williams, 1997), 17-item Hamilton Rating Scale for Depression (HRSD; Hamilton, 1960), 15-item Geriatric Depression Scale (GDS; Sheikh & Ja, 1986), informant-rated version of the Apathy Evaluation Scale (AES; Marin, Biedrzycki, &

Firinciogullari, 1991) and details of past psychiatric history. Apathy was defined as a score of $\geq$37 on the AES. Dementia diagnoses were made according to DSM-IV criteria and were confirmed by consensus at team clinical meetings.

Measures of cognitive function

Global cognitive function was assessed with the MMSE and premorbid intelligence with the National Adult Reading Test-Revised (NART-R; Nelson, 1983). A large neuropsychological test battery was also administered, although the results are not reported here (for details, see Brodaty et al., 2005; Sachdev et al., 2006). Trained psychologists had performed all tests. Neuropsychological assessments of subjects judged to be significantly depressed at clinical interview were deferred until patients improved. This was defined as either a GDS score of less than 5, a reduction in self or informant-reported symptoms and/or further psychiatric assessment of improvement.

Neuroimaging

All subjects received either a CT or MRI scan immediately after their stroke and a subset had an MRI scan at baseline approximately 3 months later. Only the baseline MRI data are presented here. MRI was performed on a 1.5 Tesla Signa GE scanner (GE Systems, Milwaukee, USA) using the following protocol: a scout mid-sagittal cut (2D, repetition time TR 300 ms, echo time TE 14 ms; 5 mm thick, number of excitations 1.5); 1.5 mm thick T1-weighted contiguous coronal sections through the whole brain using a FSPGR sequence (TR 14.3 ms, TE 5.4 ms); and 4 mm thick (0 skip) T2-weighted fluid attenuation inversion recovery (FLAIR) coronal slices through the whole brain (TR 8900, TE 145, TI 2200, FOV 25, 256×192). Trained staff scored these with good inter-rater (κ scores 0.7–0.9 on various measures) and intra-rater ($\kappa = 0.8$–0.9) reliability. ANALYZE® (Mayo Foundation, Rochester MI, USA) software was used for all ratings. Cortical atrophy was rated on a 0–2 scale in the frontal, temporal and mid-parietal regions, the sum of which gave the cortical atrophy score (higher scores = more atrophy, maximum score = 6).

Stroke characteristics determined from baseline MRI scans were number of infarcts (large infarcts and lacunes), sides of infarct (this was a clinical determination based on neuroimaging, history and clinical findings) and total volume of infarction. *Where* necessary, determination of the new lesion site was made at a consensus conference using clinical and baseline neuroimaging data.

Statistical analyses

The Statistical Package for the Social Sciences (SPSS) version 14 for Windows (SPSS Inc., Chicago, IL) was used for all analyses. Independent-samples *t*-tests were employed for between-group comparisons on continuous variables. Univariate analyses of the association between demographics in both groups were performed using chi-square analysis. For 2×2 tables, Yates' Continuity Correction is reported. Fisher's exact test was used in the analysis of 2×2 tables with expected frequencies lower than five in two or more cells. Odds ratios (with 95% confidence intervals) were calculated for the relative risk of psychiatric disorders and cerebrovascular risk factors. For all analyses, probability levels reported were two-tailed, and the level of significance was 0.05. The significance of predictors for a favourable outcome was calculated by multivariate logistic regression (predictors entered collectively if not otherwise specified).

Results

The sample

Of the 205 stroke participants recruited at baseline, there were no significant differences between the 168 who completed the baseline assessment and the 37 who did not with respect to age ($t = -1.13$, $p = 0.26$), education ($t = -1.21$, $p = 0.23$) or sex ($\chi^2 = 0.00$, $p = 0.99$). Of the 168 stroke participants followed-up, there were no significant differences between the 125 for whom the MMSE, ADL and IADL scales were completed at both time points and the 43 patients for whom they were not with respect to age ($t = 1.00$, $p = 0.32$), education ($t = 0.04$, $p = 0.97$), gender ($\chi^2 = 0.77$, $p = 0.30$), MMSE ($t = -0.23$, $p = 0.82$) or ADL + IADL score ($t = -0.70$, $p = 0.49$).

Rate of favourable outcome in the stroke sample

The 125 stroke participants who constitute the sample for this study were classified into two groups. Thirty-one had a favourable outcome at follow-up (i.e., 24.8%) whilst 94 had a poorer outcome (75.2%).

Sociodemographic and clinical comparisons of groups with favourable versus poorer outcomes

Those in the favourable outcome group were found to be significantly younger, have higher premorbid IQ scores and were less likely to have been diagnosed with apathy, but not depression, at baseline. They were also less likely to have had a cerebrovascular event in the period between assessments. There were no significant differences between groups as regards to gender, years of education, being married, or stroke characteristics such as stroke severity, side of stroke, number of strokes or total volume of infarcts (Table 1). As regards vascular risk factors, the 'doing well' group reported significantly fewer risk factors overall. Of the individual risk factors, those doing well were less likely to have been diagnosed with atrial fibrillation at baseline (3/30 [10.0%] versus 29/91 [31.9%]; OR = 0.24, 95% CI 0.08, 0.66).

Table 1. Sociodemographic and clinical comparisons of doing well and not doing well groups in the stroke ample.

	Doing well $n=31$	Not doing well $n=94$	t- or χ^2 or OR value	p or 95% CI
Demographic/clinical data				
Age	67.13 (8.19)	73.57 (8.40)	3.73	≤0.001
Sex (% Male)	20/31 (64.5)	53/94 (56.4)	0.34	0.56
Education (Years)	10.90 (3.30)	9.71 (2.33)	−1.87	0.07
NART-R IQ	108.50 (9.08)	102.90 (10.17)	−2.60	0.01
Marital status (% married)	17/30 (56.7)	49/92 (53.3)	0.01	0.91
Living with a family member (%)				
IQCODE at baseline	2.98 (0.48)	3.01 (0.60)	0.21	0.83
ADL + IADL at baseline	13.84 (0.45)	11.11 (3.46)	−7.08	≤0.001
MMSE at baseline	29.00 (1.37)	27.29 (2.55)	−4.55	≤0.001
Dementia at baseline (%)	0/31 (0.0%)	22/90 (24.4%)	9.26	0.002
Apathy at baseline (%)	3/31 (9.7)	25/77 (32.5)	OR = 0.22	0.08, 0.65
DSM major/minor Depression at baseline (%)	2/30 (6.7)	14/89 (15.7)	OR = 0.39	0.11, 1.40
HDRS score at baseline	1.95 (2.54)	3.63 (5.52)	1.35	0.18
GDS score at baseline	1.43 (1.93)	3.69 (3.12)	4.44	≤0.001
Stroke characteristics				
ESS	91.84 (12.59)	87.83 (14.94)	−1.20	0.23
No. of infarcts[a]	1.53 (1.37)	1.36 (1.37)	−0.45	0.66
Total volume infarcts[a]	597.19 (934.28)	2671.83 (7474.58)	1.14	0.26
Side of stroke (%)				
Right	18/30 (60.0)	44/93 (47.3)		
Left	10/30 (33.3)	46/93 (49.5)	2.68	0.26
Bilateral	2/30 (6.7)	3/93 (3.2)		
Total atrophy[a]	1.25 (1.81)	1.94 (1.85)	1.32	0.19
Sum of vascular risk factors at baseline	2.16 (1.07)	2.85 (1.40)	2.25	0.03
Interval cerebrovascular Event (%)	2/30 (6.7)	18/76 (23.7)	OR = 0.23	0.07, 0.79

Notes: NART-R IQ = National Adult Reading Test Revised Intelligence Quotient; IQCODE = Informant Questionnaire on Cognitive Decline in the Elderly; ADL = Activities of Daily Living; IADL = Instrumental Activities of Daily Living; ADL + IADL = sum of total score for both scales; MMSE = Mini-Mental State Examination; GDS = Geriatric Depression Scale, ESS = European Stroke Scale, AES = Apathy Evaluation Scale. Suggested Bonferroni correction for stroke characteristics: $p = 0.01$ (0.05/4).
[a]70 patients received an MRI scan.

At follow-up assessment, MMSE scores for the doing well and not doing well groups were 28.97 (0.88) and 25.99 (3.10), respectively. Likewise, ADL + IADL scores were 14.00 (0.00) and 9.88 (4.15). Significant univariate predictors (age, National Adult Reading Test Intelligent Quotient (NART IQ), sum of vascular risk factors, Geriatric Depression Scale (GDS) were tested in multivariate logistic regression. To avoid over modelling, the variables sum of vascular risk factors and GDS were tested in two different models with age and NART IQ as covariates. The first model included age, NART IQ and sum of vascular risk factors (age: OR = 0.87, 95%CI 0.80, 0.94; NART IQ: OR = 1.08, 95%CI 1.01, 1.16; sum of vascular risk factors: OR = 0.45, 95%CI 0.24, 0.85). The second model included age, NART IQ and GDS (age: OR = 0.88, 95%CI 0.82, 0.95; NART IQ: OR = 1.07, 95%CI 1.01, 1.14; GDS: OR = 0.74, 95%CI 0.55, 1.02). Since there were no people with dementia at baseline in the doing well group (Table 1), odds ratios could be calculated and so this highly significant variable could not be tested in multiple logistic regressions.

Discussion

The group that had a favourable outcome and was doing well 12–15 months after stroke (approximately one quarter of the sample) was found to be significantly younger and to have significantly higher premorbid IQ. This latter finding is consistent with the theory of brain reserve which holds that a direct relationship does not exist between degree of brain pathology and clinical outcomes (Stern, 2002). The premise of the brain reserve theory is that there is a threshold which must be crossed before impairment is observed; those with more brain reserve have to decline further before they reach this threshold (Stern, 2002). Brain reserve is believed to be conveying via increasing levels of education and/or premorbid intelligence (Tuokko, Garrett, McDowell, Silverberg, & Kristjansson, 2003). We found that higher premorbid IQ was a more useful predictor of better long-term post-stroke outcomes than education. Premorbid IQ has also been shown to be a more valid measure of reserve in late-life depression and dementia (Bhalla et al., 2005; Schmand, Smit, Geerlings, & Lindeboom, 1997).

With respect to cardiovascular risk factors, those with a more positive outcome had significantly fewer vascular risk factors at baseline, in particular less atrial fibrillation. This is broadly consistent with other reports of lower diabetes prevalence in improvers following stroke and supports the hypothesis that people tend towards an improving course after stroke except in the presence of concurrent small vessel disease (Ballard et al., 2003; Desmond et al., 1996) and/or further cerebrovascular events as also shown in this study. Atrial fibrillation (on history or as a new diagnosis) has been shown to be relatively common in people with first-ever stroke and is associated with increased mortality and disability at discharge (Paciaroni et al., 2005).

While other studies have suggested that social support is beneficial to recovery after stroke (Kwakkel et al., 1996), this was not the case in this study. Here, social support is narrowly defined as having a family member or partner living with the participant and was not a predictor of favourable outcome. There was also no significant difference between groups according to marital status.

It was found that significantly less apathy, but not less depression (DSM-IV major or minor), at baseline was associated with a favourable outcome. Apathy may have an important moderating role in outcomes following stroke or may be an indicator of more frontal and subcortical pathology (Brodaty et al., 2005). Apathy has been shown to be associated with decreased scores on the Functional Independence Measure and increased time in hospital following stroke (Galynker et al., 1997). Furthermore such subjects tend to have a poor response to rehabilitation which is out of proportion to their stroke severity and is instead attributable to their motivation difficulties (Clark & Mankikar, 1979). Apathy has also been shown to be an important predictor, beyond level of cognition, of functional dependence in vascular dementia (Zawacki et al., 2002). As might be anticipated, dementia was more likely in those with poorer outcomes and people with dementia were more likely to have cognitive and functional losses in the period between assessments.

It is interesting that there were no differences between groups according to initial stroke severity or side of stroke as determined by clinical symptoms. In the subgroup that received an MRI, there were also no differences with respect to number of infarcts, stroke volume or total atrophy. This suggests that such coarse measures are unhelpful in predicting outcome. Combined effects of specific lesion site, volume of lesion(s), pre-existing atrophy (and brain reserve) are more likely to be predictive of outcome especially when combined with clinical data. Large numbers of subjects are required to provide sufficient power for such analyses.

This study has importantly provided a better understanding of positive outcomes following stroke since those performing at ceiling were also included and also outcomes were considered functionally as well as cognitively. Several methodological issues bear comment. There is no retrospective rating of pre-stroke functional performance and so it is possible that some degree of impairment was present prior to this stroke. We included first and repeat stroke subjects in this study although given that there were no significant differences between groups according to number of strokes or previous stroke it is unlikely that this had a significant effect on the results. Limited subject numbers in the MRI sub-study precluded the examination of a large number of imaging variables. We acknowledge that it could be argued that the whole group is doing well on account of their post-stroke survival. We also acknowledge that many participants who had done poorly may have dropped out of the study, thereby underestimating their number, although those not continuing in the study at baseline had comparable cognitive and functional scores.

We conclude that among persons who have survived stroke for at least one week there is a subgroup who have reasonably good outcomes in the 12–15 months following stroke. Those with poorer outcomes are more likely to have experienced interim cerebrovascular events, to have risk factors for cerebrovascular disease and to have a diagnosis of dementia three months after their stroke. Doing well after stroke does not appear to be related to either stroke severity or number of previous strokes. Brain reserve appears to play a protective role and apathy may be an important moderating variable.

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